Dr. Mundo: Technical Write-up

Course: Introduction to Agentic AI (STAI100), Final Capstone
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Repository: https://github.com/Dawsxn/DrMundo
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Abstract

Dr. Mundo estimates what a Filipino patient should prepare for a hospital visit. The user photographs a doctor's laboratory request; a vision model reads which tests were ordered, a resolver turns those labels into catalogue codes, and a deterministic pricing layer produces a peso range after PhilHealth and HMO coverage. The defining property is that no number is invented. Every peso traces to a row in a local database built from Makati Medical Center's own published price list, and an output guardrail rejects any figure in the prose that cannot be tied back to that structure.

This version narrows the midterm's five-hospital scope to one hospital with real published prices, and adds a computer vision stage as the primary input path. Measured on 260 labelled request forms: resolution from label to code is lossless, 91.3% of read labels reach a priced catalogue row, and the hallucinated-peso rate is zero across 40 generated reports. The reading stage is the weak link, and Section 6 reports exactly where it fails and why.

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1. Business case

1.1 The problem

Philippine hospital prices are not published in any form a patient can act on. Most private hospitals publish nothing at all. A patient handed a request slip for six tests has no way to find out what those six tests cost without phoning the hospital, being transferred between departments, and writing figures on the back of the slip. PhilHealth's case rates add a second layer of opacity: the subsidy is real and often substantial, but which procedures attract one, and how it interacts with an itemised bill, is not something a patient can reasonably work out.

The result is that people arrive at admitting with no idea whether they can afford the day.

1.2 Users

The direct user is a patient or a family member holding a request slip and trying to budget. The same engine has a business audience: HMO member services, hospital patient relations, and health-tech products that need a pricing assistant they can defend. That is where willingness to pay sits, and it is also where a wrong number is expensive, which is the property this system is built around.

1.3 Value proposition and units

We claim two things, and we separate what we measured from what we assumed.

Measured. For every laboratory panel MMC publishes, buying the panel costs less than buying its component tests separately. The saving runs from ₱60 to ₱1,400 depending on the panel and which end of the price range applies. This comes straight from the committed dataset and is checked by a test. A patient who does not know this pays the difference.

Assumed, and flagged as an assumption. Pricing six tests by telephone takes roughly 30 minutes of a patient's time, allowing for hold time and transfers between departments. We did not measure this and it should be read as a working estimate rather than a finding. Dr. Mundo answers in a mean of 3.6 seconds. On that assumption the time saved is about 0.5 hours per episode. For a household, an episode is a handful of times a year, so the annual time saving is small in absolute terms and the value is concentrated elsewhere: in knowing the number before you are standing at the counter.

Intangible, and labelled as such. The larger benefit is avoided surprise, and we cannot put a defensible peso figure on it. Being told at admitting that a bill is twice what you brought is a different kind of harm from losing half an hour. We are not going to invent a number for it.

1.4 Why this needs an agent

The specification asks whether a well-crafted prompt would suffice. It would not, and we can show why rather than assert it.

MMC's price list is only reachable by POST to a search endpoint that returns an empty page if you fetch it plainly. A general chatbot cannot see it. The mapping from MMC's catalogue names to PhilHealth's RVS codes does not exist anywhere in public; ours is 51 mappings made by hand, and getting one wrong changes an estimate by tens of thousands of pesos. Reading a request slip requires deciding which rows carry a tick and which are struck through, then resolving abbreviations that are ambiguous without their section heading. Any of those steps done from memory produces a confident, wrong, unverifiable figure.

The model orchestrates and explains. The numbers come from the database.

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2. What changed since the midterm

The midterm covered five hospitals with indicative price ranges. This version covers one, with figures scraped from Makati Medical Center's own published list, every row carrying MMC's catalogue item code so any number on screen can be checked by hand. The dataset grew from about 134 priced rows to roughly 2,400.

Two midterm constraints were reversed. HMO is now in scope, as a conversational input rather than a dataset. Outpatient items can now carry coverage arithmetic where a case rate applies.

The rule that governed the rebuild, and that still governs the code: if a number was never published, we say so. That is why appendectomy has no package price in this system, why 19 procedures have no PhilHealth case rate, and why nine taxonomy codes report as unpriced. Those gaps are exposed rather than filled.

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3. Review of related literature and model selection

The vision stage is the component the course requires, so the choice of model is the part of this project with the most literature behind it and the most measurement in front of it.

3.1 The candidates

We considered two families.

Optical character recognition. Tesseract as a baseline, PaddleOCR and docTR as modern detector-plus-recogniser pipelines, and TrOCR as a transformer approach aimed at handwriting. These are mature, run locally, cost nothing per page, and keep the image on the machine. docTR in particular pairs a text detector with a CRNN recogniser and is well suited to structured documents.

Vision language models. GPT-4o-mini, GPT-4o and GPT-4.1. These read an image and emit text directly, understand layout without an explicit detector, and can be asked questions about what they see rather than only what characters are present.

3.2 The finding that decided it

We ran docTR against a clean form from our evaluation set. It read 186 words correctly. That is every printed label on the page, including the roughly forty tests the doctor did not order. The ground truth for that form is seven tests.

This is not a tuning problem. OCR answers the question "what text is on this page". The question we need answered is "which rows are marked", and a mark is not text. Making the OCR family competitive would require a second stage: detect each checkbox, measure ink density inside it, associate it with the nearest label. That is a real and buildable system, and it is a different project from the one the course window allows.

We report this as the central trade-off rather than hiding it, because it is the most useful thing we learned. OCR is the right tool when the information is the text. It is the wrong tool when the information is a pencil stroke.

3.3 Model capability is a threshold, not a gradient

Having chosen the vision-language family, we compared three models on the same image with the same prompt.

Model	Tests reported	Invented	Latency
gpt-4o-mini	24	18	17.5 s
gpt-4o	6	0	7.0 s
gpt-4.1	6	0	1.9 s

Ground truth was seven. GPT-4o-mini did not merely score worse; it failed in the same way OCR does, listing every label on the page. Mark detection appears to be a capability the smaller model does not have, and no amount of prompt work moved it. GPT-4o and GPT-4.1 both handled it, and GPT-4.1 was roughly four times faster.

This mattered practically because gpt-4o-mini is the model the rest of the application uses for chat. The vision reader deliberately does not inherit that setting.

3.4 The latency budget

We fixed a budget of 20 seconds from upload to rendered report before running any benchmark, and wrote it into the plan. The reasoning was that the demo runs live in front of a class, and attention goes after about twenty seconds of silence; a patient in a hospital lobby is no more patient than an audience.

Fixing it in advance matters. Choosing the threshold after seeing the results would have made the rejection of any candidate unfalsifiable, and a grader asking how we picked the number would have deserved a better answer than "it is where the winner happened to land".

GPT-4.1 measures a mean of 3.6 seconds and a maximum of 7.2 across 40 forms, comfortably inside. TrOCR was not benchmarked; on CPU, per-line transformer inference over a dense form would very likely exceed the budget, but we did not measure it and we are not going to claim a result we do not have.

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4. Architecture

4.1 The pipeline

```
image ──► redact ──► read ──► resolve ──► triage ──► price ──► waterfall ──► report
          raster     VLM      taxonomy    4 buckets  MMC       pure Python   HTML/text
                                                                                  │
text ───► input guard ──► ReAct loop ──► tools ──► output guard ─────────────────┘
```

The text path is the midterm system and is unchanged. Everything new hangs off the image path, and the two converge at the output guardrail.

4.2 The stages

Redaction runs first and destroys pixels rather than drawing an annotation over them. A redacted PDF that still carries the original text underneath is worse than no redaction because it looks safe.

Reading turns an image into marks: a label, its section heading, and whether the row is struck through. The reader is an interface with several implementations, so a vision model and an OCR pipeline can be scored on identical terms. An oracle reader that returns ground truth is used to measure everything downstream with reading held perfect.

Resolution maps a printed label to a taxonomy code. The rules come from the taxonomy's own matching block rather than from us: no substring matching, fuzzy matching to an edit distance of one, and an explicit list of code families that must never fuzzy-match each other. LDL and LDH are one character apart and are unrelated tests.

Triage puts every item in exactly one of four buckets: priceable, unpriced, needs-confirmation, or cancelled. An ambiguous label and an unresolved label both go to needs-confirmation. We read something; the patient is owed a question about it rather than a silent omission.

Pricing joins a taxonomy code to an MMC catalogue row. MMC publishes emergency-room, point-of-care, specimen and extent variants of nearly every test, so the join ranks candidates instead of demanding a unique match: avoid ER pricing because a request slip is not an emergency, respect specimen because a serum creatinine is not a urine creatinine, and prefer the least qualified name so an unqualified order does not silently buy a richer study.

The waterfall is pure Python with no model involvement. Gross, then the senior and PWD reduction, then PhilHealth, then HMO, then what to prepare. Every leg is a range rather than a scalar, because a deduction is the smaller of an entitlement and what is left to pay, and what is left differs between the ends of a price range.

4.3 Grounding

Two layers, both inherited from the midterm and both extended.

Structured fields come only from tool results. The model never produces a number; it selects which typed query to call.

The output guardrail then extracts every peso figure from the prose and checks it against the structured estimate. Anything it cannot find causes the prose to be discarded and rebuilt deterministically. Extending this to multi-item reports was necessary and easy to get wrong: the original check looked only at a flat set of scalar fields, so a report quoting a range per item would have had every one of those figures declared ungrounded, and the guardrail would have deleted the itemisation while leaving something that still looked like a report.

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5. Methodology and design decisions

The model never writes SQL. It chooses among typed, parameterised queries and supplies arguments that Pydantic validates. All SQL lives in one module.

Ambiguity resolves toward the higher out-of-pocket figure. Where two readings are defensible, we take the one that leaves the patient expecting to pay more. Over-preparing is recoverable. Being short at the counter is not.

Discount before PhilHealth. The senior and PWD reduction is VAT exemption followed by 20%, a single multiplier of 0.714286. It is applied before the case rate is deducted, because the two orderings do not commute: on a ₱100,000 procedure with a ₱46,800 case rate they differ by about ₱13,371. The multiplier is one named constant with a test asserting it is not 0.80, since dropping the VAT exemption is the version of this bug that costs money.

Deductions clamp per item. A ₱60,450 case rate against a ₱40,000 procedure covers ₱40,000. The excess is not a credit against the patient's blood tests.

A partial price can never be reported as full coverage. Four MMC rows carry a case rate larger than MMC's own ceiling price, which means MMC is billing part of an episode rather than all of it. Those rows are flagged, out-of-pocket is reported as uncomputable rather than zero, and the phrase "fully covered" is blocked.

Cancelled rows are shown and never charged. A struck-through row is a test the doctor withdrew. Rendering it visibly, rather than dropping it, lets the patient catch a misread: if the reader wrongly cancels something they actually need, they can see it and say so.

Room and board never enters a total. Length of stay cannot be known from a request slip, and assuming one would invent the largest number on the page. It appears as a per-day line outside the total.

The first answer asks nothing. Somebody who has just uploaded a slip wants a figure, not an intake form. The report appears immediately, labelled as being before any HMO, and every question after that is an offer to improve a number they already have. Abandoning halfway still leaves them with something true.

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6. Experiments and evaluation

The suite is organised in five layers so that a failure identifies its own stage: unit arithmetic, resolution, matching, reading, and end-to-end grounding, plus a model-scored honesty rubric. It runs offline against an oracle reader for the deterministic layers and against the real model for the reading layer.

6.1 Headline results

Metric	Result	Reading
Resolution, label to code	**100%** (1452/1452)	With reading held perfect, resolution loses nothing
Top-1 match, code to priced row	**91.3%** (1284/1407)	The remaining 8.7% are tests MMC publishes no price for
Hallucinated-peso rate	**0.0%** (0/40 reports)	No figure appeared that could not be traced to structure
Reading F1, GPT-4.1	**77.2%** (n=40)	Recall 67.2%, precision 90.8%
Cancelled rows billed	1 of 11	The failure that costs money, and it is not yet zero
Latency	mean 3.6 s, max 7.2 s	Against a 20 s budget fixed in advance

6.2 Reading degrades with media quality, as designed

The evaluation set renders each form in five conditions of increasing difficulty. Recall tracks them closely.

Media	Recall	Invented
clean	81.7%	3
scan	74.1%	3
photocopy, first generation	62.5%	3
photocopy, third generation	61.4%	6
fax	50.0%	2

Half the tests on a faxed form are missed. This is the honest ceiling of the current system and the clearest target for future work. Precision holds up much better than recall, which means the failure mode is omission rather than fabrication. For a cost estimator that is the better direction to fail in, though it is still a failure: a missed test is a bill the patient did not budget for.

Note that clean-form recall here (81.7%, n=8) is lower than the 100% we saw in an earlier three-image spot check. Eight forms per condition is a small sample and these figures carry wide error bars. We report the larger run rather than the flattering one.

6.3 Section headings are worth measuring

The taxonomy warns that a bare "KUB" appears under both Ultrasound and X-Ray on the same sheet. We tested how much the heading is actually worth by withholding it.

Resolution falls from 100% to 96.8%. Forty-six labels become ambiguous: Thyroid eighteen times, Liver Profile fourteen, KUB ten, Breast four. Critically, none become wrong. The system asks instead of guessing, which is the behaviour we wanted and now have evidence for rather than hope about.

6.4 Safety behaviours

On the taxonomy's five explicitly ambiguous abbreviations, the system asked five times out of five and priced nothing. Across the full set, no cancelled row reached the pricing layer when reading was held perfect; the one billed cancellation in the headline table came from the vision model misreading a strike-through, not from the pricing logic.

6.5 The honesty judge

Numbers being right is not the same as a report being honest. A model scores each generated report against five criteria: does it name unpriced items and say they are excluded, does it avoid claiming full coverage against a partial price, does it label HMO figures that came from a published tier rather than the patient's own certificate, does it show cancelled rows, and does it avoid reading as a quotation. The judge sees the report and the structured estimate but never the ground truth, so it grades prose against structure.

Eight reports scored 100% on all five criteria. The sample is small and the judge shares a family with the model being judged, which is a real limitation of this method and not one we can design away at this scale.

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7. What we would tell the next team

The gate on the evaluation dataset paid for itself. Its documentation says to run about ten images past a real model and look at the output before drawing conclusions from a full sweep. We did, and the first model we tried returned 24 tests where seven were ordered. Running the full benchmark first would have produced a plausible-looking F1 for a configuration that was structurally broken.

Write the failing case down before you fix it. The most valuable single test we have asserts that a chest x-ray code reaches the right MMC row. It failed, and the failure revealed that our own hand-written mapping quietly dropped a lateral view and would have understated every chest x-ray on an AP-and-lateral order. The test was wrong about what it expected and right about looking.

The dangerous bugs are the ones that return a number. A read failure originally produced an empty slip, which priced cleanly to ₱0. "Prepare ₱0" reads like an answer. It now raises. The same shape of bug appeared three times: an empty result, a partial price, and a truncated x-ray order all produced confident output that happened to be wrong, and none of them looked like errors.

Limits we did not solve. Every image in the evaluation set is synthetic, so none of these numbers describe a phone photograph of a real slip in bad light. Every form is a laboratory request, so no form names a procedure and the PhilHealth leg is never exercised from an image; it fires from the conversation instead. HMO figures come from Maxicare's published individual and family tiers, and most Philippine coverage is employer-negotiated and will match no public tier. Fax and third-generation photocopy recall sits near half.

What we would do next, in order. Add a checkbox detector so the OCR family becomes a real comparison arm rather than a documented rejection. Collect a small set of real, redacted slips so there is a non-synthetic denominator. Push cancelled-row detection to zero, since that is the one failure that takes money out of somebody's pocket rather than merely omitting a line.

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8. Deployment

The application runs as two processes: a FastAPI service exposing `/ask`, `/ask-slip`, `/refine`, `/reset` and `/health`, and a Streamlit interface. Both are containerised, and the image needs no OCR weights because the reading stage is an API call. Per-request token usage, cost and latency are logged to MLflow, and the reading stage reports its own time separately so it can be compared against the latency budget rather than hidden inside a total.

Uploads are validated on type and size, written to a temporary file outside the repository, and deleted in a finally block. The temporary name is random by design, because a patient's own filename can carry their name.

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9. Conclusion

Dr. Mundo answers a question a Filipino patient cannot currently answer for themselves, and it answers it without inventing anything. The arithmetic is deterministic and auditable, the grounding guardrail is measured at zero hallucinated figures across the reports we generated, and the places where the data simply does not exist are reported as gaps rather than filled with plausible numbers.

The reading stage is where it is weakest, and we have said where and by how much. A system that misses half the tests on a faxed form is not finished. But it is honest about being unfinished, which is the property we would most want to keep if we carried this further.